

Table 4. Steering results using ProtoMech (sequential), PLT (sequential), CAA, and selecting random mutations on seven DMS assays. All variants were constrained to a maximum of five mutations away from the wildtype.

Method	DMS	Mean score $\uparrow$	Max score $\uparrow$	Top 10% score $\uparrow$	Top 20% score $\uparrow$
ProtoMech (sequential)	SPG1_STRSG_Olson_2014	1.93 ± 0.55	3.24	2.98 ± 0.22	2.78 ± 0.25
	HIS7_YEAST_Pokusaeva_2019	1.28 ± 0.09	1.50	1.45 ± 0.03	1.41 ± 0.04
	GRB2_HUMAN_Faure_2021	-0.17 ± 0.11	0.15	0.06 ± 0.06	0.01 ± 0.07
	GFP_AEQVI_Sarkisyan_2016	3.85 ± 0.21	4.38	4.31 ± 0.06	4.20 ± 0.12
	CAPSD_AAV2S_Sinai_2021	0.37 ± 0.33	1.35	1.05 ± 0.17	0.88 ± 0.22
	RASK_HUMAN_Weng_2022_abundance	-0.04 ± 0.09	0.20	0.13 ± 0.04	0.10 ± 0.04
	A4_HUMAN_Seuma_2022	0.33 ± 0.42	1.53	1.30 ± 0.26	1.00 ± 0.36
PLT (sequential)	SPG1_STRSG_Olson_2014	1.97 ± 0.48	3.18	2.88 ± 0.20	2.67 ± 0.25
	HIS7_YEAST_Pokusaeva_2019	1.27 ± 0.06	1.41	1.38 ± 0.02	1.36 ± 0.03
	GRB2_HUMAN_Faure_2021	-0.18 ± 0.14	0.15	0.12 ± 0.05	0.03 ± 0.09
	GFP_AEQVI_Sarkisyan_2016	4.40 ± 0.22	5.15	4.92 ± 0.19	4.76 ± 0.21
	CAPSD_AAV2S_Sinai_2021	0.81 ± 0.53	2.05	1.87 ± 0.11	1.67 ± 0.22
	RASK_HUMAN_Weng_2022_abundance	-0.19 ± 0.08	0.06	-0.02 ± 0.06	-0.07 ± 0.07
	A4_HUMAN_Seuma_2022	-0.05 ± 0.38	1.03	0.67 ± 0.24	0.52 ± 0.24
CAA	SPG1_STRSG_Olson_2014	0.70 ± 0.00	0.70	0.70 ± 0.00	0.70 ± 0.00
	HIS7_YEAST_Pokusaeva_2019	0.52 ± 0.14	0.77	0.77 ± 0.00	0.71 ± 0.06
	GRB2_HUMAN_Faure_2021	-0.40 ± 0.11	-0.20	-0.21 ± 0.02	-0.23 ± 0.03
	GFP_AEQVI_Sarkisyan_2016	2.93 ± 0.23	3.16	3.16 ± 0.00	3.16 ± 0.00
	CAPSD_AAV2S_Sinai_2021	-0.26 ± 0.55	0.45	0.45 ± 0.00	0.45 ± 0.00
	RASK_HUMAN_Weng_2022_abundance	-0.35 ± 0.07	-0.28	-0.28 ± 0.00	-0.29 ± 0.01
	A4_HUMAN_Seuma_2022	-1.90 ± 0.03	-1.87	-1.87 ± 0.00	-1.87 ± 0.00
Random	SPG1_STRSG_Olson_2014	-2.57 ± 0.82	-1.15	-1.36 ± 0.16	-1.54 ± 0.21
	HIS7_YEAST_Pokusaeva_2019	0.56 ± 0.24	1.08	0.98 ± 0.06	0.91 ± 0.08
	GRB2_HUMAN_Faure_2021	-0.84 ± 0.23	-0.45	-0.49 ± 0.03	-0.53 ± 0.05
	GFP_AEQVI_Sarkisyan_2016	2.76 ± 0.46	3.72	3.51 ± 0.18	3.35 ± 0.21
	CAPSD_AAV2S_Sinai_2021	-1.02 ± 0.53	0.27	0.10 ± 0.14	-0.21 ± 0.33
	RASK_HUMAN_Weng_2022_abundance	-0.69 ± 0.25	-0.23	-0.31 ± 0.05	-0.36 ± 0.06
	A4_HUMAN_Seuma_2022	-1.41 ± 0.47	-0.43	-0.68 ± 0.21	-0.87 ± 0.25

Table 5. Steering results using ProtoMech (full replacement), PLT (sequential), CAA, and selecting random mutations on seven DMS assays. All variants were constrained to a maximum of five mutations away from the wildtype.

Method	DMS	Mean score $\uparrow$	Max score $\uparrow$	Top 10% score $\uparrow$	Top 20% score $\uparrow$
ProtoMech (full replacement)	SPG1_STRSG_Olson_2014	1.39 $\pm$ 0.68	3.18	2.67 $\pm$ 0.30	2.38 $\pm$ 0.36
	HIS7_YEAST_Pokusaeva_2019	1.14 $\pm$ 0.08	1.33	1.28 $\pm$ 0.04	1.24 $\pm$ 0.05
	GRB2_HUMAN_Faure_2021	-0.15 $\pm$ 0.13	0.13	0.08 $\pm$ 0.04	0.04 $\pm$ 0.05
	GFP_AEQVL_Sarkisyan_2016	4.64 $\pm$ 0.45	5.70	5.47 $\pm$ 0.12	5.34 $\pm$ 0.17
	CAPSD_AAV2S_Sinai_2021	1.07 $\pm$ 0.57	3.28	2.40 $\pm$ 0.47	1.94 $\pm$ 0.58
	RASK_HUMAN_Weng_2022_abundance	-0.13 $\pm$ 0.09	0.05	0.02 $\pm$ 0.01	0.00 $\pm$ 0.02
	A4_HUMAN_Seuma_2022	-0.13 $\pm$ 0.35	1.18	0.67 $\pm$ 0.30	0.41 $\pm$ 0.34
PLT (sequential)	SPG1_STRSG_Olson_2014	1.97 $\pm$ 0.48	3.18	2.88 $\pm$ 0.20	2.67 $\pm$ 0.25
	HIS7_YEAST_Pokusaeva_2019	1.27 $\pm$ 0.06	1.41	1.38 $\pm$ 0.02	1.36 $\pm$ 0.03
	GRB2_HUMAN_Faure_2021	-0.18 $\pm$ 0.14	0.15	0.12 $\pm$ 0.05	0.03 $\pm$ 0.09
	GFP_AEQVL_Sarkisyan_2016	4.40 $\pm$ 0.22	5.15	4.92 $\pm$ 0.19	4.76 $\pm$ 0.21
	CAPSD_AAV2S_Sinai_2021	0.81 $\pm$ 0.53	2.05	1.87 $\pm$ 0.11	1.67 $\pm$ 0.22
	RASK_HUMAN_Weng_2022_abundance	-0.19 $\pm$ 0.08	0.06	-0.02 $\pm$ 0.06	-0.07 $\pm$ 0.07
	A4_HUMAN_Seuma_2022	-0.05 $\pm$ 0.38	1.03	0.67 $\pm$ 0.24	0.52 $\pm$ 0.24
CAA	SPG1_STRSG_Olson_2014	0.70 $\pm$ 0.00	0.70	0.70 $\pm$ 0.00	0.70 $\pm$ 0.00
	HIS7_YEAST_Pokusaeva_2019	0.52 $\pm$ 0.14	0.77	0.77 $\pm$ 0.00	0.71 $\pm$ 0.06
	GRB2_HUMAN_Faure_2021	-0.40 $\pm$ 0.11	-0.20	-0.21 $\pm$ 0.02	-0.23 $\pm$ 0.03
	GFP_AEQVL_Sarkisyan_2016	2.93 $\pm$ 0.23	3.16	3.16 $\pm$ 0.00	3.16 $\pm$ 0.00
	CAPSD_AAV2S_Sinai_2021	-0.26 $\pm$ 0.55	0.45	0.45 $\pm$ 0.00	0.45 $\pm$ 0.00
	RASK_HUMAN_Weng_2022_abundance	-0.35 $\pm$ 0.07	-0.28	-0.28 $\pm$ 0.00	-0.29 $\pm$ 0.01
	A4_HUMAN_Seuma_2022	-1.90 $\pm$ 0.03	-1.87	-1.87 $\pm$ 0.00	-1.87 $\pm$ 0.00
Random	SPG1_STRSG_Olson_2014	-2.65 $\pm$ 1.04	-0.52	-0.85 $\pm$ 0.30	-1.20 $\pm$ 0.42
	HIS7_YEAST_Pokusaeva_2019	0.44 $\pm$ 0.33	0.86	0.83 $\pm$ 0.01	0.79 $\pm$ 0.05
	GRB2_HUMAN_Faure_2021	-0.82 $\pm$ 0.30	-0.33	-0.40 $\pm$ 0.06	-0.45 $\pm$ 0.07
	GFP_AEQVL_Sarkisyan_2016	2.79 $\pm$ 0.47	3.80	3.48 $\pm$ 0.23	3.35 $\pm$ 0.21
	CAPSD_AAV2S_Sinai_2021	-1.04 $\pm$ 0.66	0.24	0.03 $\pm$ 0.13	-0.14 $\pm$ 0.19
	RASK_HUMAN_Weng_2022_abundance	-0.67 $\pm$ 0.32	-0.04	-0.18 $\pm$ 0.12	-0.27 $\pm$ 0.13
	A4_HUMAN_Seuma_2022	-1.40 $\pm$ 0.49	-0.52	-0.78 $\pm$ 0.15	-0.90 $\pm$ 0.16

F. ProtoMech Visualization

In this section, we detail how we visualize circuits in the paper and the main functions of our visualization tool.

F.1. Visualizing Circuits

In ProtoMech, we follow the procedure according to (Ameisen et al., 2025) in order to visualize our circuits. Notably, this involves creating a *local replacement model*, which creates a unique replacement model for each sequence.

Local Replacement. The local replacement model uses a base sequence s and assumes access to all ground-truth inputs and outputs $\mathbf{x}_s^\ell, \mathbf{y}_s^\ell$ for s . Given an input sequence, the update rule becomes:

$$\mathbf{x}^\ell = \hat{\mathbf{x}}_{\text{pre}}^\ell + \sum_{\tilde{h} \in H_\ell} \tilde{h}(\mathbf{x}_{\text{pre}}^\ell), \quad (12)$$

where $\tilde{h}(\mathbf{x}_{\text{pre}}^\ell)$ represents the output of the attention head h with its attention pattern frozen to the value of the attention pattern from the base model. The denominators of the LayerNorm functions are fixed to the standard deviation of $\mathbf{x}_s^\ell$ for each ℓ . We also add the error between the MLP output $\mathbf{y}^\ell$ and the CLT approximation $\hat{\mathbf{y}}^\ell$ at each layer. By adding in the residual error at every layer, Equation (8) becomes:

$$\mathbf{x}_{\text{pre}}^{\ell+1} = \mathbf{x}^\ell + \hat{\mathbf{y}}^\ell + (\mathbf{y}_s^\ell - \hat{\mathbf{y}}_s^\ell), \quad (13)$$

where $\hat{\mathbf{y}}_s^\ell$ is the CLT reconstruction of $\mathbf{y}_s^\ell$. If the input sequence is s itself, the output is the same as the output of the base model. Some implementation details are adapted from (Krzus, 2025).

Visualizing Latents. Using the entire Swiss-Prot dataset, we identify the top-10 maximally activating sequences from Swiss-Prot in order to identify motifs or family-specific patterns that latents consistently light up (available through our ProtoMech visualizer). This complements us analyzing the activations of latents when analyzing the input sequence and allows for a more holistic view of identifying meaningful latents. To create Fig. 3, we feed in the canonical sequences associated with the UniProt IDs P83104 and Q5RKL5 for the protein kinase and NADP+ binding domains, respectively, and create the local replacement model using the circuits discovered from the sequential replacement model. We project the amino acid activations onto their Swiss-Prot structure. To create Fig. 4, we feed in the GB1 domain defined by (Olson et al., 2014) and project the amino acid activations onto the PDB structure 1pga (Gallagher et al., 1994).

Edge Weight Computation. To find the edge weight between a given source and target latent, we use the same method as detailed in Appendix E of (Ameisen et al., 2025) for computing the edge weights (also called virtual weights, which we use interchangeably) for edges between feature latents. Formally, we can write the edge weight between a source latent s and target latent t as

$$A_{s \rightarrow t} = a_s w_{s \rightarrow t} = a_s \sum_{\ell_s \leq \ell_t} (W_{\text{dec}, s}^{\ell_s \rightarrow \ell_t})^T J_{t_s, \ell_s \rightarrow t_t, \ell_t}^\nabla W_{\text{enc}, s}^{\ell_s},$$

where $J_{t_s, \ell_s \rightarrow t_t, \ell_t}^\nabla$ represents the Jacobian between token s at layer ℓ_s and token t at layer ℓ_t , with frozen attention patterns and layernorm denominators. Intuitively, $w_{s \rightarrow t}$ computes the gradient between the source activation a_s and the target pre-activation z_t , where $\mathbf{z}^{\ell_t} := \mathbf{W}_{\text{enc}}^{\ell_t} (\mathbf{x}^{\ell_t} - \mathbf{b}_{\text{pre}}^{\ell_t}) + \mathbf{b}_{\text{enc}}^{\ell_t}$.

F.2. Visualization Tool

We provide the ProtoMech visualization tool alongside our paper submission. The platform enables users to interactively explore the circuit structures presented in the paper and analyze new protein sequences. Users can either select from pre-loaded examples or upload custom circuit data by providing the following files:

1. `activation_indices.json`: Contains latents (specified by layer and latent index) that activate at each position in the sequence.
2. `seq.txt`: Contains the full protein sequence.
3. `top_activations.json`: Contains the top-activating sequences for each latent.
4. `virtual_weights.json` (optional): Contains all virtual weights between pairs of latents.